

CS 406: Numerical Methods

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References

- [Numerical Methods for Scientific and Engineering Computation by M. K. Jain, S. R. K. Iyengar, and R. K. Jain](#)
- [Advanced Engineering Mathematics by Erwin Kreyszig](#)
- [Numerical Methods for Engineers by Steven C. Chapra and Raymond P. Canale](#)
- [Numerical Methods for Engineers Course by IIT Madras](#)
- [Numerical Methods Course by IIT Roorkee](#)

Numerical Methods is a branch of mathematics that designs, analyses and implements algorithms to obtain approximate numerical solutions to complex problems that cannot be solved exactly using analytical formulae.

Finite Data Representation

Since computers have limited memory, it cannot store a number with infinite digits such as π or $\frac{1}{3}$. This introduces the fundamental concept of **Finite Data Representation**.

Fixed Point Numbers

- A method where a specific number of digits is permanently assigned to the integer part and a specific to the fractional part.
- The decimal part never moves.
- Examples:
 - $1234.6251 = 1234 + 0.6251$
 - $1234 = 1 \times 10^3 + 2 \times 10^2 + 3 \times 10^1 + 4 \times 10^0$
 - $0.6251 = 6 \times 10^{-1} + 2 \times 10^{-2} + 5 \times 10^{-3} + 1 \times 10^{-4}$

Decimal Number System

- Decimal Number System has **base 10** with digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.
- Any decimal number N can be written as
 - $N = d_{n-1}d_{n-2}\dots d_1d_0.d_{-1}d_{-2}\dots d_{-m}$
 - $(N)_{10} = d_{n-1} \times 10^{n-1} + \dots + d_1 \times 10^1 + d_0 \times 10^0 + d_{-1} \times 10^{-1} + \dots + d_{-m} \times 10^{-m}$

Binary Number System

- Binary Number System has **base 2** with digits 0, 1 called as **bits**.
- Any binary number N can be written as
 - $N = b_{n-1}b_{n-2}\dots b_1b_0.b_{-1}b_{-2}\dots b_{-m}$
 - $(N)_2 = d_{n-1} \times 2^{n-1} + \dots + d_1 \times 2^1 + d_0 \times 2^0 + d_{-1} \times 2^{-1} + \dots + d_{-m} \times 2^{-m}$
- Exercises:
 - Binary to Decimal conversion and vice versa.

Octal Number System

- Octal Number System has **base 8** with digits 0, 1, 2, 3, 4, 5, 6, 7.
- Any octal number N can be written as
 - $N = o_{n-1}o_{n-2}\dots o_1o_0.o_{-1}o_{-2}\dots o_{-m}$
 - $(N)_8 = d_{n-1} \times 8^{n-1} + \dots + d_1 \times 8^1 + d_0 \times 8^0 + d_{-1} \times 8^{-1} + \dots + d_{-m} \times 8^{-m}$
- Exercises:
 - Octal to Decimal conversion and vice versa.
 - Octal to Binary conversion and vice versa.

Hexadecimal Number System

- Hexadecimal Number System has **base 16** with digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F.
- Any hexadecimal number N can be written as
 - $N = h_{n-1}h_{n-2}\dots h_1h_0.h_{-1}h_{-2}\dots h_{-m}$
 - $(N)_{16} = d_{n-1} \times 16^{n-1} + \dots + d_1 \times 16^1 + d_0 \times 16^0 + d_{-1} \times 16^{-1} + \dots + d_{-m} \times 16^{-m}$
- Exercises:
 - Hexadecimal to Decimal conversion and vice versa.
 - Hexadecimal to Binary conversion and vice versa.
 - Hexadecimal to Octal conversion and vice versa.

Floating Point Numbers

- A method where the decimal part can float or move to adapt to the scale of the number.
- $x = \pm m \times b^e$, where m is mantissa, b is base, and e is exponent.
- Example: $1234.56 = 1.23456 \times 10^3$
- Exercise: Subtract two floating point numbers 0.36143447×10^7 and 0.36132346×10^7
 - $0.36143447 \times 10^7 - 0.36132346 \times 10^7 = 0.00011101 \times 10^7$
- **Normalized floating point numbers** are numbers whose fractional part has been shifted to remove any redundant zeroes and save space
 - Example: $0.00011101 \times 10^7 \rightarrow 0.11101 \times 10^4$
- Exercise: Add two floating point numbers 0.123×10^3 and 0.456×10^2

- $0.123 \times 10^3 \rightarrow 1.23 \times 10^2$
 - $0.123 \times 10^3 + 0.456 \times 10^2 = 1.686 \times 10^2$
 - $1.686 \times 10^2 \rightarrow 0.1686 \times 10^3$
 - $0.1686 \times 10^3 \approx 0.169 \times 10^3$ (by roundoff)
 - $0.1686 \times 10^3 \approx 0.168 \times 10^3$ (by chopping)
-

Errors

- In numerical computation, the values used in a calculation are often approximated.
- So, the error e is $e = x - \bar{x}$, where x is the **true value** and $\bar{x}$ is the **approximated value**.
- In terms of magnitude, $|e| = |x - \bar{x}|$
- The error which is machine dependent is called **machine epsilon**.

Inherent Error

It is the quantity which is already present in the statement of the problem before its solution.

The Round-off Error

It is the quantity R which must be added to the finite representation of a computed number in order to make it the true representation of that number.

The Truncation Error

It is the quantity t that must be added to the true representation of the quantity in order for the result to be exactly equal to the quantity we are seeking to generate.

The Propagation of Error

- Consider $y = f(x)$. If x contains an error, then the calculated value of y will also contain an error.
- The propagation of an error is the process by which errors present in input data or intermediate computations effect the accuracy of the final computed result.

Absolute Error

- Let $y = f(x)$ and x contain a small error dx .
- From Taylor Expansion, $f(x + dx) \approx f(x) + f'(x) \cdot dx$
- We know, $dy = f(x + dx) - f(x)$
- So, $dy \approx f'(x) \cdot dx$
- Therefore, $|dy| = |f'(x)| \cdot |dx|$
- Example: Let $y = x^2$, $x = 10$, and $x = 0.1$.
 - $dy = f'(x) \cdot dx$

- $f'(x) = 2x$
- $dy = (2 \times 10) \times 0.1$
- $dy = 2$

Relative Error

- $\left| \frac{dy}{y} \right| = \left| \frac{x \cdot f'(x)}{f(x)} \right| \cdot \left| \frac{dx}{x} \right|$
- From the previous example:
 - $\frac{x \cdot f'(x)}{f(x)} = \frac{x \cdot 2x}{x^2} = 2$
 - $\frac{dy}{y} = 2 \times \frac{dx}{x} = 2 \times \frac{0.1}{10}$
 - $\frac{dy}{y} = 0.02$
 - Approximately 2% error in x^2 .

Propagation of Error in Several Variables

- Consider, $z = f(x, y)$
- $dz = \frac{\partial f}{\partial x} \cdot dx + \frac{\partial f}{\partial y} \cdot dy$
- The maximum possible absolute error can be computed as:
 - $|dz| \approx \left| \frac{\partial f}{\partial x} \right| \cdot |dx| + \left| \frac{\partial f}{\partial y} \right| \cdot |dy|$
- Generalization, $z = f(x_1, x_2, \dots, x_n)$
 - $|dz| \approx \sum_{i=1}^n \left| \frac{\partial f}{\partial x_i} \right| \cdot |dx_i|$

Addition	Subtraction	Multiplication	Division
$z = x + y$	$z = x - y$	$z = x \cdot y$	$z = \frac{x}{y}$
$dz = dx + dy$	$dz = dx - dy$	$\frac{dz}{z} = \frac{dx}{x} + \frac{dy}{y}$	$\frac{dz}{z} = \frac{dx}{x} - \frac{dy}{y}$
Max absolute error:	Max absolute error:	Relative error:	Relative error:
$ dz \leq dx + dy $	$ dz \leq dx + dy $	$\left \frac{dz}{z} \right \leq \left \frac{dx}{x} \right + \left \frac{dy}{y} \right $	$\left \frac{dz}{z} \right \leq \left \frac{dx}{x} \right + \left \frac{dy}{y} \right $

Error Propagation Exercises

1. $y = x^3$, where $x = 2$, with error as 0.01. Find approximate and relative error.
 - Given, $y = x^3$, $x = 2$, and $dx = 0.01$.
 - $f'(x) = 3x^2$
 - $|dy| = |f'(x)| \cdot |dx| = 3x^2 \cdot dx$
 - $dy = 3 \times 2^2 \times 0.01$
 - **Approximate error:** $|dy| = 0.12$
 - $\left| \frac{dy}{y} \right| = \left| \frac{x \cdot f'(x)}{f(x)} \right| \cdot \left| \frac{dx}{x} \right|$
 - $\frac{x \cdot f'(x)}{f(x)} = \frac{x \cdot 3x^2}{x^3} = 3$
 - $\left| \frac{dy}{y} \right| = 3 \times \left| \frac{dx}{x} \right| = 3 \times \frac{0.01}{2}$

- **Relative error:** $\left|\frac{dy}{y}\right| = 0.015 = 1.5\%$
2. $z = x + y$, where $x = 10 \pm 0.1$ and $y = 20 \pm 0.2$. Find maximum error in z .
- $|dz| \leq |dx| + |dy|$
 - $|dz| \leq 0.1 + 0.2$
 - **Maximum absolute error:** $|dz| \leq 0.3$
3. $A = L \cdot W$, where $L = 20 \pm 0.1$ cm and $W = 10 \pm 0.05$ cm. Find in A, the maximum error and relative error.
- $dz = y \cdot dx + x \cdot dy$
 - $dz = 10(0.1) + 20(0.05)$
 - **Maximum error:** $dz = 2\text{cm}^2$
 - $\left|\frac{dz}{z}\right| \leq \left|\frac{dx}{x}\right| + \left|\frac{dy}{y}\right|$
 - $\left|\frac{dz}{z}\right| \leq \frac{0.1}{20} + \frac{0.05}{10}$
 - **Relative error:** $\left|\frac{dz}{z}\right| = 0.01 = 1\%$

Accuracy

It refers to how close an approximate value is to the true or exact value.

Precision

It refers to the degree of consistency or fineness of the numerical representation or measurements.

If measurements are close to one another, they are precise, but they are not necessarily accurate.

Solutions of Non-linear Equations (Finding Roots)

- Non-linear equations are equations that can be generally written as $f(x) = 0$, where $f(x)$ is non-linear in x .
- Examples:
 - $x^2 - x = 0$
 - $x^3 - x - 1 = 0$
 - $e^x - x - 2 = 0$
 - $\sin\left(x - \frac{x}{2}\right) = 0$
- **Root:** A number r is called a root or zero of $f(x)$ if $f(r) = 0$.
 - Example: $x^2 - 4 = 0$
 - For $x = 2$, $f(2) = 0$
 - For $x = -2$, $f(-2) = 0$
 - So, roots are ± 2

Geometrical Interpretation

- Let $y = f(x)$ and $f(x) = 0$. So, $y = 0$. Therefore, a root is the point where $y = f(x)$ intersects the x-axis.
- Consider $y = f(x) = x^3 - x - 1 = 0$
 - $f(1) = 1 - 1 - 1 = -1$
 - $f(2) = 8 - 2 - 1 = 5$
 - Function changes from negative to positive between 1 and 2.
 - So, we know a root exists somewhere in $1 < x < 2$
- Root finding problem:** Given $f(x)$, find $x = r$ such that $f(r) = 0$.

Types of Root Finding Methods

- Bracketing Method:** In this method, first we identify the interval $[a, b]$ such that a root is known to lie inside it.
 - Example: Bisection Method
- Open Methods:** These methods do not necessarily maintain interval containing the root.
 - Example: Newton-Raphson Method, Secant Method, Fixed Point Iteration.
 - They converge much faster but they may also fail if the initial approximation is not suitable.

Initial Approximations

- Initial approximations to the root are often known from the physical consideration of the problem. Otherwise, graphical methods are generally used to obtain initial approximations to the root.
- Another commonly used method to obtain the initial approximation to the root is based on intermediate value theorem.
- The Intermediate Value Theorem** states that if $f(x)$ is a continuous function on some interval $[a, b]$ and $f(a) \cdot f(b) < 0$, then the equation $f(x) = 0$ has at least one real root or an odd number of real roots in the interval $[a, b]$.
- Example 1:
 - The equation $8x^3 - 12x^2 - 2x + 3 = 0$ has three real roots. Find the intervals each of unit length containing each one of those roots.

x	-2	-1	0	1	2	3
f(x)	-105	15	3	-3	13	105

- From the above, we find that equation $f(x) = 0$ has roots in the intervals $(-1, 0)$, $(0, 1)$, and $(1, 2)$.
 - The exact roots are -0.5 , 0.5 , and 1.5 .

- Example 2:
 - Obtain intervals which contain the root of the equation $f(x) = \cos(x) - x \cdot e^x$

x	0	0.5	1	1.5	2
f(x)	1	0.0532	-2.1700	-6.6518	-13.1982

- - The above equation has roots in the interval (0.5, 1).
 - The exact root is 0.5177573637...

Bisection Method

This method is used on the repeated applications of the intermediate value theorem.

Pseudocode

- Given $f(x)$, a , b :
- Check $f(a) \cdot f(b) < 0$
- Repeat $c = \frac{a+b}{2}$
- If $f(c) = 0$:
 - Return c
- If $f(a) \cdot f(c) < 0$:
 - $b = c$
- Else:
 - $a = c$
- Until stopping criteria is satisfied,
- Return c

Exercises on Bisection Method

Exercise 1

Perform four iterations of the bisection method to find the approximate root of the equation $f(x) = x^3 - x - 1 = 0$.

- Initial intervals $a = 1$ and $b = 2$
- $f(1) = -1$, $f(2) = 5$
- $f(1) \cdot f(2) < 0$.
- So, root lies in the interval (1, 2).
- **Iteration 1:**
 - Midpoint, $c = \frac{1+2}{2} = 1.5$

- $f(c) = f(1.5) = (1.5)^3 - (1.5) - 1 = 0.875$
- $f(1) \cdot f(1.5) < 0$
- So, root lies in the interval $(1, 1.5)$.
- **Iteration 2:**
 - Midpoint, $c = \frac{1+1.5}{2} = 1.25$
 - $f(c) = f(1.25) = (1.25)^3 - (1.25) - 1 = -0.296875$
 - $f(1.25) \cdot f(1.5) < 0$
 - So, root lies in the interval $(1.25, 1.5)$.
- **Iteration 3:**
 - Midpoint, $c = \frac{1.25+1.5}{2} = 1.375$
 - $f(c) = f(1.375) = (1.375)^3 - (1.375) - 1 = 0.22461$
 - $f(1.25) \cdot f(1.375) < 0$
 - So, root lies in the interval $(1.25, 1.375)$.
- **Iteration 4:**
 - Midpoint, $c = \frac{1.25+1.375}{2} = 1.3125$
 - $f(c) = f(1.3125) = (1.3125)^3 - (1.3125) - 1 = -0.0515$
 - $f(1.3125) \cdot f(1.375) < 0$
 - So, root lies in the interval $(1.3125, 1.375)$.
- Approximate value of the root, $r \approx \frac{1.3125+1.375}{2} = 1.34375$

Exercise 2

Perform five iterations of the bisection method to obtain the smallest possible positive root of the equation $f(x) = x^3 - 5x + 1 = 0$.

- Initial intervals $a = 0$ and $b = 1$
- $f(0) = (0)^3 - 5(0) + 1 = 1$
- $f(1) = (1)^3 - 5(1) + 1 = -3$
- $f(0) \cdot f(1) < 0$.
- So, root lies in the interval $(0, 1)$.
- **Iteration 1:**
 - Midpoint, $c = \frac{0+1}{2} = 0.5$
 - $f(c) = f(0.5) = (0.5)^3 - 5(0.5) + 1 = -1.375$
 - $f(0) \cdot f(0.5) < 0$
 - So, root lies in the interval $(0, 0.5)$.
- **Iteration 2:**
 - Midpoint, $c = \frac{0+0.5}{2} = 0.25$
 - $f(c) = f(0.25) = (0.25)^3 - 5(0.25) + 1 = -0.234375$
 - $f(0) \cdot f(0.25) < 0$
 - So, root lies in the interval $(0, 0.25)$.
- **Iteration 3:**

- Midpoint, $c = \frac{0+0.25}{2} = 0.125$
- $f(c) = f(0.125) = (0.125)^3 - 5(0.125) + 1 = 0.376953$
- $f(0.125) \cdot f(0.25) < 0$
- So, root lies in the interval $(0.125, 0.25)$.
- **Iteration 4:**
 - Midpoint, $c = \frac{0.125+0.25}{2} = 0.1875$
 - $f(c) = f(0.1875) = (0.1875)^3 - 5(0.1875) + 1 = 0.069092$
 - $f(0.1875) \cdot f(0.25) < 0$
 - So, root lies in the interval $(0.1875, 0.25)$.
- **Iteration 5:**
 - Midpoint, $c = \frac{0.1875+0.25}{2} = 0.21875$
 - $f(c) = f(0.21875) = (0.21875)^3 - 5(0.21875) + 1 = -0.083282$
 - $f(0.1875) \cdot f(0.21875) < 0$
 - So, root lies in the interval $(0.1875, 0.21875)$.
- Approximate value of the root, $r \approx \frac{0.1875+0.21875}{2} = 0.203125$

Exercise 3

Perform five iterations of the bisection method to find the approximate root of the equation $f(x) = \cos(x) - x \cdot e^x = 0$.

- Initial interval $a = 0$ and $b = 1$
- $f(0) = \cos(0) - (0)e^0 = 1$
- $f(1) = \cos(1) - (1)e^1 = 0.540302 - 2.718282 = -2.177980$
- $f(0) \cdot f(1) < 0$.
- So, root lies in the interval $(0, 1)$.
- **Iteration 1:**
 - Midpoint, $c = \frac{0+1}{2} = 0.5$
 - $f(c) = f(0.5) = \cos(0.5) - (0.5)e^{0.5} = 0.877583 - 0.824361 = 0.053222$
 - $f(0.5) \cdot f(1) < 0$
 - So, root lies in the interval $(0.5, 1)$.
- **Iteration 2:**
 - Midpoint, $c = \frac{0.5+1}{2} = 0.75$
 - $f(c) = f(0.75) = \cos(0.75) - (0.75)e^{0.75} = 0.731689 - 1.587750 = -0.856061$
 - $f(0.5) \cdot f(0.75) < 0$
 - So, root lies in the interval $(0.5, 0.75)$.
- **Iteration 3:**
 - Midpoint, $c = \frac{0.5+0.75}{2} = 0.625$
 - $f(c) = f(0.625) = \cos(0.625) - (0.625)e^{0.625} = 0.810963 - 1.167654 = -0.356691$
 - $f(0.5) \cdot f(0.625) < 0$
 - So, root lies in the interval $(0.5, 0.625)$.

- **Iteration 4:**

- Midpoint, $c = \frac{0.5+0.625}{2} = 0.5625$
- $f(c) = f(0.5625) = \cos(0.5625) - (0.5625)e^{0.5625} = 0.845924 - 0.987218 = -0.141294$
- $f(0.5) \cdot f(0.5625) < 0$
- So, root lies in the interval $(0.5, 0.5625)$.

- **Iteration 5:**

- Midpoint, $c = \frac{0.5+0.5625}{2} = 0.53125$
 - $f(c) = f(0.53125) = \cos(0.53125) - (0.53125)e^{0.53125} = 0.862176 - 0.903686 = -0.041510$
 - $f(0.5) \cdot f(0.53125) < 0$
 - So, root lies in the interval $(0.5, 0.53125)$.
- Approximate value of the root, $r \approx \frac{0.5+0.53125}{2} = 0.515625$

Error Bound for Bisection Method

- How accurate is the Bisection Method?
- For the initial interval (a, b) , the interval length is $b - a$.
- After one iteration, the interval length becomes $\frac{b-a}{2}$.
- After two iterations, it is $\frac{b-a}{2^2}$.
- After n iterations, it is $\frac{b-a}{2^n}$.
- This gives us the direct error bound.
- Since the root lies inside any $(\pm, \pm)$ interval, the maximum distance between the midpoint and the root is half the interval length.
- Therefore, after n iterations, $|x_n - r| \leq \frac{b-a}{2^{n+1}}$, where x_n is the approximate root and r is the actual root.

Finding the Error Bound

- Let $a = 1$ and $b = 2$. Find the error bound after five iterations.
- $|x_n - r| \leq \frac{b-a}{2^{n+1}}$
- $|x_n - r| \leq \frac{2-1}{2^6}$
- $|x_n - r| \leq \frac{1}{64}$
- $|x_n - r| \leq 0.015625$

Number of Iterations Required

- Let us assume that we want an error bound, $|x_n - r| \leq \epsilon$.
- So, we require $\frac{b-a}{2^{n+1}} \leq \epsilon$.
- $2^{n+1} \geq \frac{b-a}{\epsilon}$
- $n + 1 \geq \frac{\log(\frac{b-a}{\epsilon})}{\log(2)}$
- So, $n \geq \frac{\log(\frac{b-a}{\epsilon})}{\log(2)} - 1$.

Priori Error in Bisection Method

- **Priori Error** is an error estimate defined before or independently of the actual approximation.
- For bisection method, $|x_n - r| \leq \frac{b-a}{2^{n+1}}$

Posteriori Error in Bisection Method

- **Posteriori Error** is an estimate based on the current numerical results.
- For bisection method, if the current interval is (a_n, b_n) , then $|x_n - r| \leq \frac{b_n - a_n}{2}$.
- This tells us how accurate our current approximation is.

Convergence of Bisection Method

- Bisection is a convergent method under standard assumptions that:
 1. $f(x)$ is a continuous function between the interval (a, b) .
 2. $f(a) \cdot f(b) < 0$.
 - Therefore, $\lim_{n \rightarrow \infty} (b_n - a_n) = 0$
 - Bisection method has guaranteed convergence under the continuity and sign change conditions.
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Secant Method

- If $f(x)$ is a first degree equation in x , then it can be readily solved. The iteration method which will produce the exact results whenever function of $f(x) = 0$ is a first degree equation is given by:
 - $f(x) = a_0x + a_1 = 0$
 - $x = -\frac{a_1}{a_0}$
- If x_{k-1} and x_k are two approximations to the root, then we determine a_0 and a_1 by using the following equations:
 - $f_{k-1} = a_0x_{k-1} + a_1$
 - $f_k = a_0x_k + a_1$
 - Where, $f_{k-1} = f(x_{k-1})$ and $f_k = f(x_k)$.
 - $\therefore a_0 = \frac{f_k - f_{k-1}}{x_k - x_{k-1}}$ and $a_1 = \frac{x_k \cdot f_{k-1} - x_{k-1} \cdot f_k}{x_k - x_{k-1}}$
- The two approximations are:
 - $x_{k-1} \rightarrow f(x_{k-1})$
 - $x_k \rightarrow f(x_k)$
- Now, instead of taking the midpoint, draw a straight line using the following two points $(x_{k-1}, f(x_{k-1}))$ and $(x_k, f(x_k))$.
- The point where this line intersects the x-axis is taken as the next approximation.

Derivation of the Secant Formula:

- The slope between the two points $(x_{k-1}, f(x_{k-1}))$ and $(x_k, f(x_k))$ is given by
$$m = \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}.$$
- The equation of the line through $(x_k, f(x_k))$ is given by $y - f(x_k) = m(x - x_k).$
- $y - f(x_k) = \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}(x - x_k)$
- At the root of the secant line, $y = 0.$
- $\therefore -f(x_k) = \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}(x - x_k)$
- $x - x_k = \frac{-f(x_k)(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})}$
- $x_{k+1} = x_k - \frac{(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})} \cdot f(x_k)$

Advantages of the Secant Method

- Does not require derivative $f'(x).$
- Usually much faster than bisection method.
- Uses function values more intelligently than bisection method.
- Easy to implement.

Limitations of the Secant Method

- Convergence is not guaranteed for arbitrary initial guesses.
- The approximation can move away from the root.
- It may encounter division by a very small quantity if $f(x_k) - f(x_{k-1})$ is approximately zero.

If the approximations are such that $f_k \cdot f_{k-1} < 0$, then the other method is known as **Regular Falsi Method**.

Exercises on Secant and Regular Falsi Methods

The real root of the equation $f(x) = x^3 - 5x + 1 = 0$ lies in the interval $(0, 1)$. Perform four iterations of the Secant Method and the Regular Falsi Method to obtain the root.

Using Secant Method

- Let $x_0 = 0$ and $x_1 = 1.$
- $f(x_0) = f(0) = 1$
- $f(x_1) = f(1) = -3$
- **Iteration 1:**
 - $x_2 = x_1 - \frac{(x_1 - x_0)}{f(x_1) - f(x_0)} \cdot f(x_1)$
 - $x_2 = 1 - \frac{(1-0)}{-3-1}(-3)$
 - $x_2 = x_1 - \frac{(x_1 - x_0)}{f(x_1) - f(x_0)} \cdot f(x_1)$
 - $x_2 = 1 - \frac{(1-0)}{-3-1}(-3)$
 - $x_2 = 1 - 0.75 = 0.25$
 - $f(x_2) = f(0.25) = (0.25)^3 - 5(0.25) + 1 = -0.234375$

- **Iteration 2:**

- $x_3 = x_2 - \frac{(x_2 - x_1)}{f(x_2) - f(x_1)} \cdot f(x_2)$
- $x_3 = 0.25 - \frac{(0.25 - 1)}{-0.234375 - (-3)}(-0.234375)$
- $x_3 = 0.25 - 0.063560 = 0.186440$
- $f(x_3) = f(0.186440) = (0.186440)^3 - 5(0.186440) + 1 = 0.074281$

- **Iteration 3:**

- $x_4 = x_3 - \frac{(x_3 - x_2)}{f(x_3) - f(x_2)} \cdot f(x_3)$
- $x_4 = 0.186440 - \frac{(0.186440 - 0.25)}{0.074281 - (-0.234375)}(0.074281)$
- $x_4 = 0.186440 + 0.015296 = 0.201736$
- $f(x_4) = f(0.201736) = (0.201736)^3 - 5(0.201736) + 1 = -0.000470$

- **Iteration 4:**

- $x_5 = x_4 - \frac{(x_4 - x_3)}{f(x_4) - f(x_3)} \cdot f(x_4)$
- $x_5 = 0.201736 - \frac{(0.201736 - 0.186440)}{-0.000470 - 0.074281}(-0.000470)$
- $x_5 = 0.201736 - 0.000096 = 0.201640$

- Approximate value of the root, $r \approx 0.201640$

Using Regula Falsi Method

- Let $x_0 = 0$ and $x_1 = 1$.
- $f(x_0) = 1, f(x_1) = -3$
- $f(x_0) \cdot f(x_1) < 0$.
- So, root lies in the interval $(0, 1)$.

- **Iteration 1:**

- $x_2 = x_1 - \frac{(x_1 - x_0)}{f(x_1) - f(x_0)} \cdot f(x_1)$
- $x_2 = 1 - \frac{(1 - 0)}{-3 - 1}(-3) = 0.25$
- $f(x_2) = f(0.25) = -0.234375$
- $f(x_0) \cdot f(x_2) < 0$
- So, root lies in the interval $(0, 0.25)$.

- **Iteration 2:**

- $x_3 = x_2 - \frac{(x_2 - x_0)}{f(x_2) - f(x_0)} \cdot f(x_2)$
- $x_3 = 0.25 - \frac{(0.25 - 0)}{-0.234375 - 1}(-0.234375)$
- $x_3 = 0.25 - 0.047468 = 0.202532$
- $f(x_3) = f(0.202532) = -0.004352$
- $f(x_0) \cdot f(x_3) < 0$
- So, root lies in the interval $(0, 0.202532)$.

- **Iteration 3:**

- $x_4 = x_3 - \frac{(x_3 - x_0)}{f(x_3) - f(x_0)} \cdot f(x_3)$
- $x_4 = 0.202532 - \frac{(0.202532 - 0)}{-0.004352 - 1}(-0.004352)$
- $x_4 = 0.202532 - 0.000877 = 0.201655$

- $f(x_4) = f(0.201655) = -0.000074$
 - $f(x_0) \cdot f(x_4) < 0$
 - So, root lies in the interval $(0, 0.201655)$.
 - **Iteration 4:**
 - $x_5 = x_4 - \frac{(x_4 - x_0)}{f(x_4) - f(x_0)} \cdot f(x_4)$
 - $x_5 = 0.201655 - \frac{(0.201655 - 0)}{-0.000074 - 1}(-0.000074)$
 - $x_5 = 0.201655 - 0.000015 = 0.201640$.
 - Approximate value of the root, $r \approx 0.201640$
-

Newton-Raphson Method

Instead of using a second line secant through the two points, Newton-Raphson uses the tangent line at the current point to estimate where the function crosses the x-axis.

- Let us consider a current approximation x_k .
- Instead of considering the entire curve, i.e., $y = f(x)$, we draw its tangent line.
- The point on the curve is $(x_k, f(x_k))$.
- The slope of the tangent at this point is given by, $f'(x_k)$.
- So, the equation of the tangent line is given by, $y - f(x_k) = f'(x_k) \cdot (x - x_k)$.
- At the root of the tangent line, $y = 0$.
- $\therefore -f(x_k) = f'(x_k) \cdot (x - x_k)$
- $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$, where x_k is the initial approximation and x_{k+1} is the new approximation.

Convergence of Newton-Raphson Method

- Newton-Raphson is a fast method, but it is not globally guaranteed to converge for initial guesses.
- Its excellent convergence behaviour is mainly a local property near a suitable root.
- A poor initial sum can cause the method to diverge, oscillate, move towards another root, or maybe encounter a point where the derivative of f , $f'(x) = 0$ make very large jumps.

Advantages of Newton-Raphson Method

- Very fast near the root.
- Usually requires fewer iterations.
- Simple iteration formula.
- Does not require bracketing intervals like bisection or regular falsi methods.
- Useful for many non-linear problems.

Limitations of Newton-Raphson Method

- Derivative is required.
- Poor initial guess can cause failure.
- Derivative can become zero or very small.
- Multiple roots can reduce the usual conversions rates.

Exercises on Newton-Raphson Method

Exercise 1

Let $f(x) = x^3 - x - 1 = 0$. Assume an initial approximation of $x_0 = 1.5$. Perform four iterations using Newton-Raphson Method.

- $f'(x) = 3x^2 - 1$
- Initial approximation, $x_0 = 1.5$
- **Iteration 1:**
 - $f(x_0) = f(1.5) = (1.5)^3 - 1.5 - 1 = 0.875$
 - $f'(x_0) = f'(1.5) = 3(1.5)^2 - 1 = 5.75$
 - $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1.5 - \frac{0.875}{5.75}$
 - $x_1 = 1.347826$
- **Iteration 2:**
 - $f(x_1) = f(1.347826) = (1.347826)^3 - 1.347826 - 1 = 0.100361$
 - $f'(x_1) = f'(1.347826) = 3(1.347826)^2 - 1 = 4.449896$
 - $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 1.347826 - \frac{0.100361}{4.449896}$
 - $x_2 = 1.325271$
- **Iteration 3:**
 - $f(x_2) = f(1.325271) = (1.325271)^3 - 1.325271 - 1 = 0.002536$
 - $f'(x_2) = f'(1.325271) = 3(1.325271)^2 - 1 = 4.269023$
 - $x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 1.325271 - \frac{0.002536}{4.269023}$
 - $x_3 = 1.324677$
- **Iteration 4:**
 - $f(x_3) = f(1.324677) = (1.324677)^3 - 1.324677 - 1 = -0.000178$
 - $f'(x_3) = f'(1.324677) = 3(1.324677)^2 - 1 = 4.264301$
 - $x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 1.324677 - \frac{-0.000178}{4.264301}$
 - $x_4 = 1.324719$
- Approximate value of the root, $r \approx 1.324719$

Exercise 2

Let $f(x) = x^3 - 5x + 1 = 0$. Assume an initial approximation of $x_0 = 0.5$. Perform four iterations using Newton-Raphson Method.

- $f'(x) = 3x^2 - 5$

- Initial approximation, $x_0 = 0.5$
- **Iteration 1:**
 - $f(x_0) = f(0.5) = (0.5)^3 - 5(0.5) + 1 = -1.375$
 - $f'(x_0) = f'(0.5) = 3(0.5)^2 - 5 = -4.25$
 - $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0.5 - \frac{-1.375}{-4.25}$
 - $x_1 = 0.176471$
- **Iteration 2:**
 - $f(x_1) = f(0.176471) = (0.176471)^3 - 5(0.176471) + 1 = 0.123143$
 - $f'(x_1) = f'(0.176471) = 3(0.176471)^2 - 5 = -4.906574$
 - $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 0.176471 - \frac{0.123143}{-4.906574}$
 - $x_2 = 0.201567$
- **Iteration 3:**
 - $f(x_2) = f(0.201567) = (0.201567)^3 - 5(0.201567) + 1 = 0.000357$
 - $f'(x_2) = f'(0.201567) = 3(0.201567)^2 - 5 = -4.878113$
 - $x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 0.201567 - \frac{0.000357}{-4.878113}$
 - $x_3 = 0.201640$
- **Iteration 4:**
 - $f(x_3) = f(0.201640) = (0.201640)^3 - 5(0.201640) + 1 = -0.000002$
 - $f'(x_3) = f'(0.201640) = 3(0.201640)^2 - 5 = -4.878023$
 - $x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 0.201640 - \frac{-0.000002}{-4.878023}$
 - $x_4 = 0.201640$
- Approximate value of the root, $r \approx 0.201640$

Exercise 3

Approximate the value of $(17)^{\frac{1}{3}}$. Assume an initial approximation of $x_0 = 2$. Perform four iterations using Newton-Raphson Method.

- Let $(17)^{\frac{1}{3}} = x$. So, $f(x) = x^3 - 17 = 0$.
- $f'(x) = 3x^2$
- Initial approximation, $x_0 = 2$
- **Iteration 1:**
 - $f(x_0) = f(2) = (2)^3 - 17 = -9$
 - $f'(x_0) = f'(2) = 3(2)^2 = 12$
 - $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 2 - \frac{-9}{12}$
 - $x_1 = 2.75$
- **Iteration 2:**
 - $f(x_1) = f(2.75) = (2.75)^3 - 17 = 3.796875$
 - $f'(x_1) = f'(2.75) = 3(2.75)^2 = 22.6875$
 - $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 2.75 - \frac{3.796875}{22.6875}$

- $x_2 = 2.582645$
- **Iteration 3:**
 - $f(x_2) = f(2.582645) = (2.582645)^3 - 17 = 0.226380$
 - $f'(x_2) = f'(2.582645) = 3(2.582645)^2 = 20.010159$
 - $x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 2.582645 - \frac{0.226380}{20.010159}$
 - $x_3 = 2.571332$
- **Iteration 4:**
 - $f(x_3) = f(2.571332) = (2.571332)^3 - 17 = 0.000997$
 - $f'(x_3) = f'(2.571332) = 3(2.571332)^2 = 19.835241$
 - $x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 2.571332 - \frac{0.000997}{19.835241}$
 - $x_4 = 2.571282$
- Approximate value of $(17)^{1/3} \approx 2.571282$

Exercise 4 - Matrix Iteration

Obtain the complex roots of the equation $f(z) = z^3 + 1 = 0$ correct to eight decimal places. Use the initial approximation of the root as $(x_0, y_0) = (0.25, 0.25)$. Compare the exact values of the roots which is $(1 \pm \frac{i\sqrt{3}}{2})$.

- Substituting $z = x + iy$ in the equation:
 - $(x + iy)^3 + 1 = (x^3 - 3xy^2 + 1) + i(3x^2y - y^3) = 0$
 - Here, $U(x, y) = x^3 - 3xy^2 + 1$ and $V(x, y) = 3x^2y - y^3$
- Formulating the Jacobian Matrix:
 - $J = \begin{bmatrix} U_x & U_y \\ V_x & V_y \end{bmatrix} = \begin{bmatrix} \frac{\delta U}{\delta x} & \frac{\delta U}{\delta y} \\ \frac{\delta V}{\delta x} & \frac{\delta V}{\delta y} \end{bmatrix}$
 - Here, $\frac{\delta U}{\delta x} = 3x^2 - 3y^2$, $\frac{\delta U}{\delta y} = -6xy$, $\frac{\delta V}{\delta x} = 6xy$, and $\frac{\delta V}{\delta y} = 3x^2 - 3y^2$.
 - $\therefore J = \begin{bmatrix} 3x^2 - 3y^2 & -6xy \\ 6xy & 3x^2 - 3y^2 \end{bmatrix}$
- Determinant of J :
 - $D_J = (3x^2 - 3y^2)(3x^2 - 3y^2) + 36x^2y^2$
 - $D_J = 9(x^2 - y^2)^2 + 36x^2y^2$
 - $D_J = 9(x^2 + y^2)^2$
- Inverse of J :
 - $J^{-1} = \frac{1}{D_J} \text{Adj}$
 - $J^{-1} = \frac{1}{9(x^2 + y^2)^2} \begin{bmatrix} 3x^2 - 3y^2 & 6xy \\ -6xy & 3x^2 - 3y^2 \end{bmatrix}$
- Newton-Raphson Matrix Iteration Formula:
 - $\begin{bmatrix} x_{k+1} \\ y_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \end{bmatrix} - J^{-1} \begin{bmatrix} U(x_k, y_k) \\ V(x_k, y_k) \end{bmatrix}$
 - $\begin{bmatrix} x_{k+1} \\ y_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \end{bmatrix} - \frac{1}{9(x_k^2 + y_k^2)^2} \begin{bmatrix} 3x_k^2 - 3y_k^2 & 6x_k y_k \\ -6x_k y_k & 3x_k^2 - 3y_k^2 \end{bmatrix} \begin{bmatrix} x_k^3 - 3x_k y_k^2 + 1 \\ 3x_k^2 y_k - y_k^3 \end{bmatrix}$

- **Iteration 1:**

- $D_0 = 9(0.25^2 + 0.25^2)^2 = 9(0.125)^2 = 0.140625$
- $\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 0.25 \\ 0.25 \end{bmatrix} - \frac{1}{0.140625} \begin{bmatrix} 0 & 0.375 \\ -0.375 & 0 \end{bmatrix} \begin{bmatrix} 0.96875 \\ 0.03125 \end{bmatrix}$
- $\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 0.16666667 \\ 2.83333333 \end{bmatrix}$

- **Iteration 2:**

- $D_1 = 9(x_1^2 + y_1^2)^2 = 9(0.16666667^2 + 2.83333333^2)^2 = 584.02777778$
- $U(x_1, y_1) = -3.00925926, \quad V(x_1, y_1) = -22.50925926$
- $\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0.16666667 \\ 2.83333333 \end{bmatrix} - \frac{1}{584.02777778} \begin{bmatrix} -24 & 2.83333333 \\ -2.83333333 & -24 \end{bmatrix} \begin{bmatrix} -3.00925926 \\ -22.50925926 \end{bmatrix}$
- $\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0.15220505 \\ 1.89374026 \end{bmatrix}$

- **Iteration 3:**

- $D_2 = 9(x_2^2 + y_2^2)^2 = 9(0.15220505^2 + 1.89374026^2)^2 = 117.25111943$
- $U(x_2, y_2) = -0.63401099, \quad V(x_2, y_2) = -6.65981678$
- $\begin{bmatrix} x_3 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0.15220505 \\ 1.89374026 \end{bmatrix} - \frac{1}{117.25111943} \begin{bmatrix} -10.68925735 & 1.72942095 \\ -1.72942095 & -10.68925735 \end{bmatrix} \begin{bmatrix} -0.63401099 \\ -6.65981678 \end{bmatrix}$
- $\begin{bmatrix} x_3 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0.19263553 \\ 1.27724322 \end{bmatrix}$

- **Iteration 4:**

- $D_3 = 9(x_3^2 + y_3^2)^2 = 9(0.19263553^2 + 1.27724322^2)^2 = 25.05378984$
- $U(x_3, y_3) = 0.06438032, \quad V(x_3, y_3) = -1.94144150$
- $\begin{bmatrix} x_4 \\ y_4 \end{bmatrix} = \begin{bmatrix} 0.19263553 \\ 1.27724322 \end{bmatrix} - \frac{1}{25.05378984} \begin{bmatrix} -4.78272540 & 1.47625459 \\ -1.47625459 & -4.78272540 \end{bmatrix} \begin{bmatrix} 0.06438032 \\ -1.94144150 \end{bmatrix}$
- $\begin{bmatrix} x_4 \\ y_4 \end{bmatrix} = \begin{bmatrix} 0.31932197 \\ 0.91041889 \end{bmatrix}$

Summary of Root-Finding Methods

Feature	Bisection	Secant	Regular Falsi	Newton-Raphson
Type	Bracketing	Open	Bracketing	Open
Basic idea	Midpoint	Secant line	Secant + Bracket	Tangent line
Initial values	(a, b)	(x_0, x_1)	(a, b)	(x_0)
Sign Change required	Yes	No	Yes	No
Derivative required	No	No	No	Yes
Maintains bracket	Yes	No	Yes	No

Feature	Bisection	Secant	Regular Falsi	Newton-Raphson
Convergence guarantee	Under standard conditions	No	Under standard conditions	No, globally
Speed near root	Slow	Fast	Moderate/variable	Very fast
Approx order	(1)	(1.618)	Variable/ typically linear	(2) locally

Final Summary of Root-Finding Methods

Method	Bracket?	Derivative?	Convergence	Main Strength
Bisection	Yes	No	Linear	Reliability
Regular Falsi	Yes	No	Generally linear	Reliability + Interpolation
Secant	No	No	Superlinear, ($p \approx 1.618$)	Speed without derivative
Newton-Raphson	No	Yes	Quadratic, ($p = 2$) locally	Very fast convergence

Muller's Method

Instead of fitting a straight line, it fits a quadratic, polynomial through three points.

- Lets consider a function $f(x) = a_0x^2 + a_1x + a_2 = 0$, which is a polynomial of degree two.
- a_0 , a_1 , and a_2 are three arbitrary parameters to be determined by prescribing three approximate conditions $f(x)$ and/or its derivatives.
- If x_{k-2} , x_{k-1} , and x_k are three approximations to the root of $f(x) = 0$, then we may determine a_0 , a_1 , and a_2 by using the condition as:
 - $f_{k-2} = a_0x_{k-2}^2 + a_1x_{k-2} + a_2$
 - $f_{k-1} = a_0x_{k-1}^2 + a_1x_{k-1} + a_2$
 - $f_k = a_0x_k^2 + a_1x_k + a_2$
- Determining a_0 , a_1 , and a_2 from the equation,

$$\begin{vmatrix} f(x) & x^2 & x & 1 \\ f_{k-2} & x_{k-2}^2 & x_{k-2} & 1 \\ f_{k-1} & x_{k-1}^2 & x_{k-1} & 1 \\ f_k & x_k^2 & x_k & 1 \end{vmatrix}$$

- $f(x) = \frac{(x-x_{k-1})(x-x_k)}{(x_{k-2}-x_{k-1})(x_{k-2}-x_k)} \cdot f_{k-2} + \frac{(x-x_{k-2})(x-x_k)}{(x_{k-1}-x_{k-2})(x_{k-1}-x_k)} \cdot f_{k-1} + \frac{(x-x_{k-2})(x-x_{k-1})}{(x_k-x_{k-2})(x_k-x_{k-1})} \cdot f_k = 0$
- When, $h = x - x_k$, $h_k = x_k - x_{k-1}$, and $h_{k-1} = x_{k-1} - x_{k-2}$,
- $\frac{h(h+h_k)}{h_{k-1}(h_{k-1}+h_k)} \cdot f_{k-2} - \frac{h(h+h_k+h_{k-1})}{h_k \cdot h_{k-1}} \cdot f_{k-1} + \frac{(h+h_k)(h+h_k+h_{k-1})}{h_k(h_k+h_{k-1})} \cdot f_k = 0$
- $\lambda = \frac{h}{h_k}$, $\lambda_k = \frac{h_k}{h_{k-1}}$, and $\partial_k = 1 + \lambda_k$
- $C_k \lambda^2 + g_k \lambda + \partial_k f_k = 0$,
- Where, $g_k = \lambda_k^2 f_{k-2} - \partial_k^2 f_{k-1} + (\lambda_k + \partial_k) f_k$
- And, $C_k = \lambda_k(\lambda_k f_{k-2} - \partial_k f_{k-1} + f_k)$
- $\lambda = \frac{-g_k \pm \sqrt{g_k^2 - 4\partial_k f_k C_k}}{2C_k}$
- Or, $\lambda = \frac{-2\partial_k f_k}{g_k \pm \sqrt{g_k^2 - 4\partial_k f_k C_k}}$
- Choosing the sign in the denominator to maximize magnitude.
 - If $g_2 < 0 \implies$ minus sign
 - If $g_2 > 0 \implies$ plus sign
- $\lambda_{k+1} = \frac{h}{h_k} = \frac{x-x_k}{x_k-x_{k-1}}$
- $x_{k+1} = x_k + (x_k - x_{k-1})\lambda_{k+1}$
- Or, $x_{k+1} = x_k + h_k \lambda_{k+1}$

Exercise 1

Perform three iterations of the Muller's Method to find the smallest positive root of the equation $f(x) = x^3 - 5x + 1 = 0$

- The smallest root lies in the interval $(0, 1)$.
- Let $x_0 = 0$, $x_1 = 0.5$, and $x_2 = 1$.
- So, $f_0 = 1$, $f_1 = -1.375$, and $f_2 = -3$.
- **Iteration 1:**
 - $h_1 = x_1 - x_0 = 0.5$, $h_2 = x_2 - x_1 = 0.5$
 - $\lambda_2 = \frac{h_2}{h_1} = \frac{0.5}{0.5} = 1$
 - $\partial_2 = 1 + \lambda_2 = 1 + 1 = 2$
 - $g_2 = 1^2(1) - 2^2(-1.375) + (1 + 2)(-3) = -2.5$
 - $C_2 = 1(1(1) - 2(-1.375) + (-3)) = 0.75$
 - Since, $g_2 < 0$:
 - $\lambda_3 = \frac{-2(2)(-3)}{-2.5 - \sqrt{(-2.5)^2 - 4(2)(-3)(0.75)}}$
 - $\lambda_3 = \frac{12}{-2.5 - \sqrt{24.25}} \approx -1.616286$
 - $x_3 = x_2 + h_2 \lambda_3 = 1 + 0.5(-1.616286) = 0.191857$
 - $f_3 = f(x_3) = (0.191857)^3 - 5(0.191857) + 1 \approx 0.047777$
- **Iteration 2:**
 - $h_2 = 0.5$, $h_3 = x_3 - x_2 = -0.808143$
 - $\lambda_3 = \frac{h_3}{h_2} = \frac{-0.808143}{0.5} = -1.616286$
 - $\partial_3 = 1 + \lambda_3 = 1 - 1.616286 = -0.616286$

- $g_3 = (-1.616286)^2(-1.375) - (-0.616286)^2(-3) + (-1.616286 - 0.616286)(0.047777) \approx -$
- $C_3 = -1.616286((-1.616286)(-1.375) - (-0.616286)(-3) + 0.047777) \approx -0.680961$
- Since, $g_3 < 0$:
- $\lambda_4 = \frac{-2(-0.616286)(0.047777)}{-2.559261 - \sqrt{(-2.559261)^2 - 4(-0.616286)(0.047777)(-0.680961)}}$
- $\lambda_4 \approx -0.011540$
- $x_4 = x_3 + h_3\lambda_4 = 0.191857 + (-0.808143)(-0.011540) = 0.201183$
- $f_4 = f(x_4) = (0.201183)^3 - 5(0.201183) + 1 \approx 0.002230$
- **Iteration 3:**
 - $h_3 = -0.808143, h_4 = x_4 - x_3 = 0.009326$
 - $\lambda_4 = \frac{h_4}{h_3} = \frac{0.009326}{-0.808143} = -0.011540$
 - $\partial_4 = 1 + \lambda_4 = 1 - 0.011540 = 0.988460$
 - $g_4 = (-0.011540)^2(-3) - (0.988460)^2(0.047777) + (-0.011540 + 0.988460)(0.002230) \approx -$
 - $C_4 = -0.011540((-0.011540)(-3) - (0.988460)(0.047777) + 0.002230) \approx 0.000120$
 - Since, $g_4 < 0$:
 - $\lambda_5 = \frac{-2(0.988460)(0.002230)}{-0.044902 - \sqrt{(-0.044902)^2 - 4(0.988460)(0.002230)(0.000120)}}$
 - $\lambda_5 \approx 0.049093$
 - $x_5 = x_4 + h_4\lambda_5 = 0.201183 + (0.009326)(0.049093) = 0.201641$
- Approximate value of root, $r \approx 0.201641$

Advantages of Muller's Method

- Does not require derivative.
- Uses quadratic approximations.
- Can find complex roots.
- Does not require a bracketing interval.
- Can be applied to general non-linear equations.

Limitations of Muller's Method

- Requires three initial approximations.
- More computationally complicated.
- Numerical issues can occur.
- No general global convergence guaranteed.

Interpolation

Let us assume we know some values of function $(x_0, y_0), (x_1, y_1), (x_2, y_2) \dots$. But we do not know the value of that function at some point lying in between the known points.

Interpolation is the process of estimating that unknown value.

- **Definition:** Interpolation is the process of estimating the value of a function at an intermediate point using known data points.
- **Need for Interpolation:**
 - In practical numerical problems, we frequently have tabulated or measured data rather than an explicit mathematical function.
 - For example: Temperature, population data, experimental observations, vehicle speeds at different times, sensor measurements, travel time data, etc.

Interpolation estimates a value *inside* the range of known data points. Example: $x = 10$ and $x = 30$ are given, find $x = 20$.

Extrapolation estimates a value *outside* the known range. Example: $x = 10$ and $x = 30$ are given, find $x = 60$.

Polynomial Interpolation

- Suppose we have ' $n + 1$ ' data points, $(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots (x_n, y_n)$.
- Based on that we can construct a polynomial of degree at most n :
 - $P_n(x) = a_0 + a_1x + a_2x^2 + \dots a_nx^n$
 - Such that, $P_n(x_i) = y_i \forall i = 0, 1, 2, \dots n$.
- Example:
 - A polynomial of degree at most 1: $P_1(x) = a_0 + a_1x$.
 - A polynomial of degree at most 2: $P_2(x) = a_0 + a_1x + a_2x^2$.

Exercise: Two Point Interpolation

Let $f(1) = 2$ and $f(3) = 6$. Find $f(2)$ using linear interpolation.

- We know, $P_1(x) = a + bx$
- $P(1) = 2, \therefore a + b = 2$
- $P(3) = 6, \therefore a + 3b = 6$
- Subtracting them, $2b = 4, \therefore b = 2$
- Substituting b we get, $a = 0$

Linear Interpolation

- For two data points (x_0, y_0) and (x_1, y_1) , the linear interpolation formula is:
 - $y = y_0 + \frac{x - x_0}{x_1 - x_0}(y_1 - y_0)$

Larange Interpolation

- Suppose we have $(n + 1)$ data points, $(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots (x_n, y_n)$.
- The Larange Interpolation Formula:

- $P_n(x) = \sum_{i=0}^n y_i L_i(x)$
- Where, $L_i(x) = \prod_{j=0, j \neq i}^n \frac{x-x_j}{x_i-x_j}$
- Example: For three points (x_0, y_0) , (x_1, y_1) , and (x_2, y_2) .
 - $L_0 = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}$
 - $L_1 = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}$
 - $L_2 = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}$
 - $P_2(x) = y_0 L_0(x) + y_1 L_1(x) + y_2 L_2(x)$

Exercises

Exercise 1

x	1	2	4
y	1	4	16

Find $f(3)$ using Linear Interpolation.

- $L_0 = \frac{(3-2)(3-4)}{(1-2)(1-4)} = -\frac{1}{3}$
- $L_1 = \frac{(3-1)(3-4)}{(2-1)(2-4)} = 1$
- $L_2 = \frac{(3-1)(3-2)}{(4-1)(4-2)} = \frac{1}{3}$
- $P_2(3) = 1 \times -\frac{1}{3} + 4 \times 1 + 16 \times \frac{1}{3} = \frac{27}{3}$
- $\therefore f(3) = 9$

Advantages of Lrange Interpolation

- Simple direct formula.
- No need to solve simultaneous equations for the coefficients.
- Works with unequally spaced data points.
- Easy to use when only one interpolation value is required.

Limitations of Lrange Interpolation

- Formula becomes lengthy when many points are used.
- If a new data point is added, the polynomial generally has to be reconstructed.
- Repeated calculations can become cumbersome.
- For many data points, Newton divided differences are often more convenient.

Newton Forward Interpolation

- Suppose the values of the data points are $x_0, x_1, x_2, \dots, x_n$ and

$x_1 - x_0 = x_2 - x_1 = x_3 - x_2 = \dots x_n - x_{n-1} = h$, then the data is called **equally spaced**.

x	0	1	2	3	4	5	6
$f(x)$	1	4	6	10	15	17	22

- Therefore, $x_i = x_0 + ih \forall i = 0, 1, 2, \dots n$.
-

Forward Difference Operator

- The forward difference operator is represented by Δ , and $\Delta f(x) = f(x + h) - f(x)$.
- For a set of equally spaced data $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$:

x	x_0	x_1	x_2	x_3
y	y_0	y_1	y_2	y_3

- **First forward difference:** $\Delta y_i = y_{i+1} - y_i$
 - $\Delta y_0 = y_1 - y_0$
 - $\Delta y_1 = y_2 - y_1$
 - $\Delta y_2 = y_3 - y_2$
- **Second forward difference:** $\Delta^2 y_i = \Delta y_{i+1} - \Delta y_i$
 - $\Delta^2 y_0 = \Delta y_1 - \Delta y_0$
- **Third forward difference:** $\Delta^3 y_i = \Delta^2 y_{i+1} - \Delta^2 y_i$
 - $\Delta^3 y_0 = \Delta^2 y_1 - \Delta^2 y_0$

Building a Difference Table

x	0	1	2	3	4
y	1	2	5	10	17

- **1st forward differences:**
 - $\Delta y_0 = 2 - 1 = 1$
 - $\Delta y_1 = 5 - 2 = 3$
 - $\Delta y_2 = 10 - 5 = 5$
 - $\Delta y_3 = 17 - 10 = 7$
- **2nd forward differences:**
 - $\Delta^2 y_0 = 3 - 1 = 2$
 - $\Delta^2 y_1 = 5 - 3 = 2$
 - $\Delta^2 y_2 = 7 - 5 = 2$
- **3rd forward differences:**
 - $\Delta^3 y_0 = 2 - 2 = 0$

- $\Delta^3 y_1 = 2 - 2 = 0$
- **4th forward difference:**
 - $\Delta^4 y_0 = 0 - 0 = 0$

Final difference table:

n	y	Δy	$\Delta^2 y$	$\Delta^3 y$
0	1	1	2	0
1	2	3	2	0
2	5	5	2	
3	10	7		
4	17			

- Constant 1st differences indicate a polynomial of degree 1, that means a linear polynomial.
- Constant 2nd differences indicate a polynomial of degree 2, that means a quadratic polynomial.
- Constant 3rd differences indicate a polynomial of degree 3, that means a cubic polynomial.
- In this example, since $\Delta^2 y$ is constant at 2, the underlying function is exactly quadratic.

Newton Forward Interpolation Formula (N.F.I.F)

- Suppose the knows data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ with equal spacing h :
 - $p = \frac{x - x_0}{h}$
- Then the **N.F.I.F** is:

$$f(x) \approx P_n(x) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots + \frac{p(p-1)(p-2) \dots (p-n+1)}{n!} \Delta^n y_0$$

Exercises

Exercise 1

Find $f(1.5)$ using N.F Interpolation, using the table above where $x_0 = 0$ and $h = 1$.

- $p = \frac{x - x_0}{h} = \frac{1.5 - 0}{1} = 1.5$
- $f(1.5) \approx P_n(1.5) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0$
- $f(1.5) = 1 + (1.5)(1) + \frac{(1.5)(0.5)}{2}(2)$
- $f(1.5) = 1 + 1.5 + 0.75 = 3.25$
- $\therefore f(1.5) = 3.25$

Exercise 2

x	10	20	30	40	50
y	100	400	900	1600	2500

Find $f(25)$ using N.F.I method.

- $x = 25, x_0 = 10, h = 10 \Rightarrow p = \frac{25 - 10}{10} = 1.5$
- $\Delta y_0 = 300, \Delta^2 y_0 = 200$ (from the difference table; $\Delta^3 y_0 = 0$ since $y = x^2$ is exactly quadratic)
- $f(25) \approx 100 + (1.5)(300) + \frac{(1.5)(0.5)}{2}(200)$
- $f(25) = 100 + 450 + 75 = 625$
- $\therefore f(25) = 625$

Exercise 3

x	1	2	3	4
y	1	8	27	64

Find $f(2.5)$ using Newton Forward Interpolation.

- $x_0 = 1, h = 1 \Rightarrow p = \frac{2.5 - 1}{1} = 1.5$
- Difference table: $\Delta y_0 = 7, \Delta^2 y_0 = 12, \Delta^3 y_0 = 6$
- $P_3(2.5) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0$
- $P_3(2.5) = 1 + (1.5)(7) + \frac{(1.5)(0.5)}{2}(12) + \frac{(1.5)(0.5)(-0.5)}{6}(6)$
- $P_3(2.5) = 1 + 10.5 + 4.5 - 0.375 = 15.625$
- $\therefore f(2.5) = 15.625$

Exercise 4

x	10	15	20	25
$f(x)$	100	225	400	625

Find $f(12)$ using Newton Forward Interpolation.

- $x_0 = 10, h = 5 \Rightarrow p = \frac{12 - 10}{5} = 0.4$
- Difference table: $\Delta y_0 = 125, \Delta^2 y_0 = 50, \Delta^3 y_0 = 0$
- $P_3(12) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0$
- $P_3(12) = 100 + (0.4)(125) + \frac{(0.4)(-0.6)}{2}(50) + 0$
- $P_3(12) = 100 + 50 - 6 = 144$

- $\therefore f(12) = 144$

Why Newton Forward Interpolation is used when the required value is near the beginning of the equally spaced table:

Newton Forward Interpolation builds its correction terms from x_0 (the **first** point in the table), using $p = \frac{x - x_0}{h}$.

When x is close to x_0 , p is small (typically $|p| < 1$), which makes each successive term in the series — $\frac{p(p-1)}{2!} \Delta^2 y_0$, $\frac{p(p-1)(p-2)}{3!} \Delta^3 y_0$, etc. — rapidly shrink in magnitude, so the series converges quickly and only a few terms are needed for good accuracy.

If x were instead near the *end* of the table, p would be large, forcing many terms to be included for the same accuracy — which is why **Newton Backward Interpolation** (built from x_n , the last point, with an analogous small parameter) is preferred in that case instead.
